

O'RUME DOMINIC URIRIE

AI / ML Engineer | LLM pipelines, autonomous agents and RAG systems in production

CONTACT

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SKILLS

AI & ML Systems

Agentic orchestration (LangChain, LangGraph, CrewAI), RAG, tool use, multi-step reasoning, prompt engineering, local LLMs (Ollama), anomaly detection, LLM evaluation

Languages & Frameworks

Python (production-grade), TypeScript, SQL, FastAPI, Flask, Next.js, React

Machine Learning

scikit-learn, pandas, PyTorch, classification and forecasting, model evaluation, experimental design, statistical testing

Cloud & DevOps

Docker, GitHub Actions (CI/CD), GitLab CI, Google Cloud Run, Vercel, Fly.io, AWS, Supabase, PyPI packaging, automated testing

Data, Observability & Security

Elastic Stack, OCR pipelines, SHA-256 audit chains, on-chain anchoring (Polygon), Daml, RBAC, STRIDE threat modelling

CERTIFICATIONS

AI Apps & Agents Developer,
Microsoft Azure

AI Governance & Responsible AI
Leveraging AI for Governance,
Regulation and Compliance
Python, University of Michigan
CCNA and CCNP, Cisco

PROFILE

AI/ML engineer building and deploying production-grade LLM pipelines, autonomous agents and RAG systems, with a focus on moving prototypes into scalable, cloud-hosted architectures. Comfortable across the whole path: raw requirements, model and pipeline design, deployment, and the monitoring that keeps a system honest once it is live. My MSc instrument for measuring where AI coding agents fail is open source and published on PyPI; the companion ledger work is catalogued in the Canton Network's official developer hub.

WORK EXPERIENCE

MCKI Solutions

AI Engineer, Enterprise Applications

November 2025 to present · Birmingham, UK

- Architect and deploy enterprise AI applications from raw requirements through to cloud-hosted production systems.
- Built a high-accuracy OCR system (98% precision) automating tax calculations and compliance review.
- Developed an AI Student Gateway mapping applicant preferences to universities and visa pathways, reducing manual support queries by 40%.
- Streamlined deployment on Google Cloud Run, holding a consistent 90+ Lighthouse score across web applications.

Simera Global

AI Liaison and Advisor

February 2026 to present · London, UK

- Advise on the technical design of AI workflows using CrewAI and LangGraph, covering secure cross-border data handling and verification layers for autonomous agents.

Vorem

AI / ML Engineer

September 2017 to October 2025

- Moved with the field from classical ML pipelines (scikit-learn, pandas) to LLM and agentic systems.
- Engineered a five-model LLM analytics workflow on Google Cloud Run, optimising for uptime, latency and cost.
- Built a quantitative FX and crypto signal system on a nine-module architecture, with API delivery channels and a governance kill-switch.
- Served as technical project supervisor, leading the development and testing of AI tools across prompt engineering and generative AI.
- Monitored production AI systems, using Elastic observability for anomaly detection and tightening IT security processes.

EDUCATION

MSc Artificial Intelligence & Business Strategy

Aston University, UK · 2026

Focus on ML pipelines and AI coding-agent quality measurement. ML coursework using UK road-accident and cardiovascular data for classification and predictive modelling.

BEng Electrical & Electronic Engineering

Covenant University · 2014

Final project: a 5 kVA AI-based changeover inverter. Industrial internship at a 1,020 MW thermal power plant.

PROJECTS

AI Code Quality Auditor: open-source measurement instrument

Python · published on PyPI · github.com/dominicrume/ai-code-quality-auditor

A Python evaluation instrument measuring where AI coding agents fail, across specification hallucination, cyclomatic complexity, duplication and security-vulnerability density, over 600 controlled measurements with the analysers blinded by interface design. Published on PyPI with over 3,600 downloads, shipped as a reusable GitHub Action with a CI gate and a JSON decision record, and served by a live public dashboard.

Tenant Hub: enterprise HMO management platform

Next.js · Supabase · MCKI Solutions

A SaaS platform supporting 100+ tenants across multiple housing brands, with role-based access control and SHA-256 audit-chain records so every action is attributable and tamper-evident. The automation saves the operator roughly 100 hours per week and about £80,000 per year.

AgenticOS: blockchain verification layer

FastAPI · Docker · Polygon · VeritaPort Ltd

Engineered the cryptographic audit-trail layer for an institutional decision-capture platform: server-side signed record stamping with on-chain anchoring, so any third party can verify record integrity in seconds without per-user wallets.

Know Your AgenticAI: spend mandates and hash-chained receipts

Daml · Python · catalogued in the official Canton Developer Hub

Enforces an AI agent's spending mandate inside a Daml choice body (cap, per-period limit, allow-list, expiry, revocation) and seals every attempt into a hash-chained receipt, including the ones the ledger refused. Three pull requests merged by the Canton Foundation's developer-relations team.

Tax & Statement Analyser

RAG and OCR · Google Cloud Run

Extracts transactions from unstructured bank statements, classifies tax categories and flags anomalies, grounding retrieval against regulatory documents for consistent compliance review.

AI Job Search Agent

Claude API · Playwright · Flask

An autonomous pipeline that discovers roles and drafts tailored applications, routed through a human-in-the-loop dashboard so nothing is submitted without explicit review. Agentic autonomy under strict governance.

Local 7B LLM Email Processor

Ollama · runs entirely on-device

Evaluates incoming email locally, deletes spam, leaves normal messages unread and flags receipts for a paperless pipeline. A daily driver, with no data leaving the machine.

RESEARCH & RECOGNITION

- **UK patent application GB2611754.9** (filed May 2026, pending) for a framework for verifiable AI accountability and provenance.
- **ACM Europe Seasonal School on Responsible AI**, Aston University, September 2026. Selected through a scored international application round that invited only the top 30–50 candidates. My team was awarded second prize in the school's project competition, for human-in-the-loop decision support in regulated markets.
- **FrontierTechX**, University of Birmingham, March 2026. Panellist, "Agentic AI: From Copilots to Autonomous Enterprises".
- **MSc dissertation** validated by two independent raters (Cohen's $\kappa = 0.870$ rater-to-rater), with the erratum that overturned one of its own cleanest findings published rather than quietly corrected.

REFERENCES

Contact details for both are available on request.

MCKI Solutions

Employer, November 2025 to present

Vorem

Employer, 2017 to 2025